

VIJAYANAND THANGAVELU

Senior Backend Engineer · Go · Python · Microservices

vijayanand.thangavelu@gmail.com | [linkedin.com/in/vijathanga](https://www.linkedin.com/in/vijathanga) | github.com/vijathanga | vijathanga.github.io/resume | Remote

SUMMARY

Senior software engineer with 10+ years building secure, large-scale backend systems for Cisco, Visa, Shopee and Rampnow. Works mainly in Go and Python, designing clean REST APIs, microservices and the CI/CD pipelines that ship them. Has also built internal engineering tools — a VS Code extension, a GoLand plugin and a golangci-lint static-analysis platform — and shipped a cross-platform C/C++ client used by 80M+ users at Cisco.

EXPERIENCE

Co-founder & CTO — Rampnow

Feb 2024 – Present · Remote

- Own the end-to-end technology function of a Web3 fintech platform: backend, APIs, security and infrastructure.
- Deliver the platform on Go microservices and TypeScript (React/Next.js), backed by PostgreSQL, Kubernetes, AWS and Cloudflare.
- Design clean, well-contracted REST APIs across the platform's services, consumed by web and mobile clients.
- Build and operate CI/CD on Jenkins; own versioning, releases and backwards compatibility across services.
- Lead a team of six (four engineers): hiring, mentoring, code-quality standards and architecture.

Freelance Software Developer — Self-employed

Jun 2022 – Feb 2024

- Lead developer for a Zero-Trust Network Access Control solution: Python control plane, REST APIs and SDN-based enforcement.
- Built cloud-orchestration automation in Go on AWS EKS; wrote Go automation scripts for an e-commerce client's order management.
- Delivered custom ERP solutions for clients in the manufacturing industry.

Web Backend Engineer — Shopee (Sea Group)

Jun 2020 – Jun 2022 · Singapore

- Lead engineer of the Orders Return team on an e-commerce platform serving 100M+ users; owned the end-to-end return lifecycle including financial settlement.
- Led migration of the Returns application from a Python monolith to Go microservices sustaining 10,000 QPS.
- Built a GoLand IDE plugin for non-prod deployments and a golangci-lint-based static-analysis system for the Orders org.
- Mentored interns, drove code reviews and contributed to architecture planning across the Orders organisation.

Senior Software Engineer — Visa Inc.

Jan 2019 – Jun 2020 · Singapore

- Built and operated CI/CD pipelines for the entire Data Product Development org within a PCI-DSS-regulated payments network.
- Built "CI/CD-as-a-Service" — a Python/Flask self-service platform for teams to create and manage their own pipelines.
- Designed a template-driven pipeline-config system driven by a single repository settings file.
- Built a Visual Studio Code extension to parse and validate the pipeline-config settings file.

Software Developer — Cisco Systems

Jul 2015 – Jul 2017 · Bangalore, India

- C/C++ developer on AnyConnect ISEPosture, a cross-platform Network Access Control client deployed to 80M+ enterprise users.
- Key contributor to next-gen posture discovery — the mechanism by which the client locates the NAC server inside an enterprise network.
- Shipped native cross-platform builds for macOS and Windows; won the Cisco Innovator Award (2016) for a Node.js productivity tool.
- Earlier intern (Jan–Jul 2015): tested distributed-storage tech for Cisco ISE; built a Syslog/Riemann log-collection PoC (HBase, CouchDB, Java).

Graduate Research Assistant — NUS-Singtel Cyber Security R&D Lab

Feb 2018 – Nov 2018 · Singapore

- Built a distributed framework for NFV deployment on network edges; implemented IoT device fingerprinting and anomaly detection with ML.
- Output: two IEEE peer-reviewed publications and a filed patent. Stack: Python, Java, SDN, Docker.

SKILLS

Languages: Go (expert), Python (expert), C/C++ (advanced), JavaScript/TypeScript (advanced), Java, SQL

Backend & APIs: Node.js, Flask, REST API design, microservices

Cloud & Infra: AWS (EKS, EC2, S3, Lambda), Kubernetes, Docker, Cloudflare

CI/CD: Jenkins, GitHub Actions, GitLab CI, infrastructure-as-code, release & versioning

Databases: PostgreSQL, MySQL, HBase, CouchDB

Security: Network Access Control, SDN, secure SDLC

EDUCATION

M.Sc., Electrical & Computer Engineering (CS specialisation)

National University of Singapore · 2017–2018 · CGPA 4.7/5.0

B.E., Electronics & Communication Engineering

PSG College of Technology · 2011–2015 · CGPA 9.8/10 · Dept. Gold Medalist

PUBLICATIONS, PATENT & AWARDS

DEFT: A Distributed IoT Fingerprinting Technique — IEEE Internet of Things Journal, Feb 2019.

NETRA: Enhancing IoT Security using NFV-based Edge Traffic Analysis — IEEE Sensors Journal, Feb 2019.

Patent (filed Jul 2018): Identification of IoT Devices Based on a Distributed Fingerprinting Solution.

Awards: Cisco Innovator Award (2016); ECE Department Gold Medalist & All-Rounder Award, PSG College of Technology (2015).